

CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Incorporating Functional Lumen Imaging Probe Into Esophageal Clinical Practice: Expert Review



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DESCRIPTION: The purpose of this American Gastroenterological Association (AGA) Institute Clinical Practice Update (CPU) is to summarize the available evidence and offer expert best practice advice on the incorporation of the functional lumen imaging probe (FLIP) into clinical practice, specifically its utility in the evaluation of esophageal symptoms, esophageal motor dysfunction, gastroesophageal reflux disease, and eosinophilic esophagitis. **METHODS:** This CPU Expert Review was commissioned and approved by the AGA Institute Governing Board and CPU Committee to provide timely guidance on a topic of high clinical importance to the AGA membership. This CPU Expert Review underwent internal peer review by the CPU Committee and external peer review through the standard procedures of *Gastroenterology*. These best practice advice statements were developed based on review of the published literature and expert opinion, and approved by the AGA Institute Governing Board. Because formal systematic reviews were not performed, these best practice advice statements do not carry formal ratings of the quality of evidence or strength of the presented considerations.

BEST PRACTICE ADVICE STATEMENTS

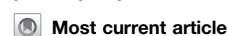
BEST PRACTICE ADVICE 1: Clinicians should perform a high-quality upper endoscopy evaluating for esophageal pathology immediately prior to considering FLIP. **BEST PRACTICE ADVICE 2:** Clinicians should consider performing FLIP if alternate investigations are inconclusive in patients with symptoms of esophageal obstruction (ie, dysphagia, esophageal-type regurgitation, and/or meal-related chest pain), and may consider FLIP as part of index endoscopy for symptomatic patients when the procedure and interpretation expertise are readily available. **BEST PRACTICE ADVICE 3:** Clinicians may use common sedation options for endoscopy (ie, propofol, fentanyl, and/or midazolam), as these medications have not been demonstrated to have a clinically significant impact on FLIP findings. **BEST PRACTICE ADVICE 4:** Normal esophagogastric junction opening on FLIP has a high negative predictive value for disorders of esophagogastric junction outflow obstruction on high-resolution manometry. **BEST PRACTICE ADVICE 5:** If upper endoscopy demonstrates findings strongly supportive of achalasia, clinicians may use FLIP findings to diagnose achalasia. **BEST PRACTICE ADVICE 6:** If upper endoscopy does not demonstrate findings supportive of achalasia, abnormal esophagogastric junction opening on FLIP should prompt further diagnostic testing. **BEST PRACTICE ADVICE 7:** Clinicians may consider performing FLIP if the

mechanism of persisting dysphagia in treated eosinophilic esophagitis is not apparent on high-quality upper endoscopy or histopathology. **BEST PRACTICE ADVICE 8:** Clinicians should not use FLIP findings to diagnose gastroesophageal reflux disease or to determine the necessity for antireflux intervention. **BEST PRACTICE ADVICE 9:** When available, clinicians should consider performing FLIP intraprocedurally during myotomy (per-oral endoscopic myotomy or laparoscopic Heller myotomy) to guide adequacy of LES disruption. **BEST PRACTICE ADVICE 10:** Clinicians may consider performing FLIP in the evaluation of persisting symptoms of esophageal obstruction (ie, dysphagia, esophageal-type regurgitation, and/or meal-related chest pain) after treatment of achalasia spectrum disorders. **BEST PRACTICE ADVICE 11:** Clinicians may consider performing FLIP in the evaluation of symptoms of esophageal obstruction after invasive foregut intervention (ie, antireflux or bariatric intervention).

Keywords: Achalasia; Dysphagia; Endoscopy; Eosinophilic Esophagitis; Esophagogastric Junction; Per-Oral Endoscopic Myotomy.

The functional lumen imaging probe (FLIP) is a useful diagnostic tool that measures esophageal dimensions and pressures in real time.^{1–5} FLIP panometry is an advancement that assesses both esophagogastric junction (EGJ) function and esophageal body contractile response. This American Gastroenterological Association Clinical Practice Update (CPU) summarizes evidence on the clinical utility of FLIP in esophageal evaluation, with particular emphasis on data published since the original American Gastroenterological Association FLIP CPU in 2017,⁶ to offer clinically relevant best practice advice on its performance and use.

Abbreviations used in this paper: AET, acid exposure time; AGA, American Gastroenterological Association; ARS, antireflux surgery; CARS, content, anatomy, resistance, and stasis; DI, distensibility index; EGJ, esophagogastric junction; EGJOO, esophagogastric junction outflow obstruction; EoE, eosinophilic esophagitis; FLIP, functional lumen imaging probe; GERD, gastroesophageal reflux disease; HRM, high-resolution manometry; IEO, inconclusive EGJ opening; LES, lower esophageal sphincter; LHM, laparoscopic Heller myotomy; NEO, normal EGJ opening; NPV, negative predictive value; POEM, per-oral endoscopic myotomy; PPI, proton pump inhibitor; REO, reduced EGJ opening.



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FLIP Protocol and Interpretation

FLIP uses impedance planimetry to evaluate the cross-sectional area (displayed as diameter), distensibility of the EGJ and/or esophageal body, and esophageal body contractile response to volumetric distention.⁷ The equipment consists of a FLIP catheter with impedance planimetry electrodes and a pressure transducer housed inside a polyurethane bag, as well as a digital data acquisition and display system. An 8-cm catheter (325N; 16 electrode pairs spaced 0.5 cm apart) primarily measures EGJ metrics, and a 16-cm catheter (322N; 16 electrode pairs spaced 1 cm apart) measures both EGJ metrics and esophageal body contractile response.¹

FLIP is performed in the sedated patient after a high-quality upper endoscopy in the left lateral position. The FLIP catheter is calibrated, referenced to atmospheric pressure, and positioned transorally with 1–3 impedance sensor pairs distal to the EGJ (16-cm catheter) or with the EGJ at the balloon center (8-cm catheter), confirmed by identifying an EGJ “waist” on the real-time display at 30–40 mL or 20-mL fill volumes, respectively.¹ If resistance is encountered during positioning, the catheter should be withdrawn and repositioned to avoid damaging the distal pressure transducer. Endoscopic visualization may be helpful in patients with abnormal anatomy (eg, esophageal diverticulum or hiatus hernia) or prior foregut surgery (eg, Roux-en-Y gastric bypass).

A FLIP panometry study involves sequential saline infusion into the FLIP bag to 50 mL, 60 mL, and 70 mL, holding each fill volume for at least 30 seconds while maintaining appropriate FLIP catheter position.¹ Key metrics include the EGJ distensibility index (DI; ie, ratio of cross-sectional area to bag pressure) and associated pressure measured at the 60-mL fill volume, the maximum EGJ diameter at the 70-mL fill volume, and the esophageal body contractile pattern based on the aggregate of the 50-mL, 60-mL, and 70-mL fill volumes.¹ When using the 8-cm catheter, similar stepwise FLIP filling is performed to 30 mL, 40 mL, and 50 mL, evaluating EGJ diameter, pressure, and EGJ-DI at each fill volume, and recording metrics from the 40-mL fill volume.² After completion, the FLIP bag is emptied and the catheter is removed.

The EGJ opening is described using 2 parameters, EGJ-DI and maximum EGJ diameter, as normal EGJ opening (NEO), reduced EGJ opening (REO), or inconclusive EGJ opening (IEO) (Table 1).¹ Esophageal body contractile response patterns can be normal (multiple distinct anterograde contractions and/or repetitive anterograde contractions) or abnormal (absent, diminished, disordered, or spastic contractile response patterns) (Figure 1).¹ FLIP pressure measurements discriminate diminished (≤ 40 mm Hg) from disordered (> 40 mm Hg) contractility when distinct anterograde contractions or spastic contractile response are not seen.^{1,8–10}

FLIP in the Evaluation of Esophageal Symptoms

Best Practice Advice 1: Clinicians should perform a high-quality upper endoscopy evaluating for esophageal pathology immediately prior to considering FLIP.

A high-quality upper endoscopy carefully evaluates for structural and mucosal abnormalities, with higher yield for erosive reflux changes or eosinophilic esophagitis (EoE) if proton pump inhibitors (PPIs) are withheld for 2–4 weeks.¹¹ The bulk of available data relate to FLIP use in evaluation of the EGJ and tubular esophagus; FLIP is contraindicated in patients with actively bleeding varices. Esophageal strictures, rings, mass lesions, diverticula, mucosal changes of EoE, erosive esophagitis, Barrett’s esophagus, and spastic changes should be noted, with biopsies obtained when appropriate.¹² A careful retroflexion after gastric air insufflation evaluates for hiatus hernia (reported using Hill or American Foregut Society grade), for abnormalities in the gastric cardia,^{13,14} and for postoperative foregut anatomy, including integrity and location of antireflux surgery (ARS), when present,¹⁵ because these can impact FLIP catheter placement. When no mucosal or structural mechanism is found to explain symptoms, the endoscope is withdrawn and the FLIP catheter is positioned. Ultimately, FLIP findings should be interpreted as part of the global clinical impression, in the context of presenting symptoms, endoscopy findings, and relevant complementary test results (Figure 2).¹

Best Practice Advice 2: Clinicians should consider performing FLIP if alternate investigations are inconclusive in patients with symptoms of esophageal obstruction (ie, dysphagia, esophageal-type regurgitation, and/or meal-related chest pain), and may consider FLIP as part of index endoscopy for symptomatic patients when the procedure and interpretation expertise are readily available.

Neither high-resolution manometry (HRM) nor timed barium esophagram (TBE) nor FLIP represents a “gold standard” for the diagnosis of esophageal motility disorders in isolation, and inconclusive findings of uncertain clinical relevance may be encountered. For instance, EGJ outflow obstruction (EGJOO) on HRM is considered inconclusive by the Chicago Classification version 4.0, and FLIP or TBE is advised to confirm or refute the clinical significance of this HRM finding.^{16–18} Furthermore, achalasia remains possible despite normal integrated relaxation pressure on HRM, and FLIP can identify obstructive EGJ metrics if clinical suspicion is high or if there is HRM-TBE discordance.^{19,20}

Discordance between esophageal test findings is not unusual in patients with symptoms of esophageal obstruction. In a study of 126 symptomatic patients who underwent HRM, FLIP, and TBE, concordance among all 3 tests assessing lower esophageal sphincter (LES) obstruction was only 57%.²¹ FLIP can sometimes demonstrate LES dysfunction and/or abnormal secondary peristalsis in symptomatic patients with no motor diagnosis on HRM.^{22,23} When tests are discordant, concordant findings on more than 1 testing modality increase confidence in the diagnosis of actionable obstructive motility disorders prior to pursuing irreversible LES disruption, whereas conservative treatments should be used if a confident achalasia spectrum diagnosis cannot be made.^{16,24}

Table 1. Use of FLIP Findings in Making Obstructive Structural and Motor Diagnoses

Diagnosis	EGJ metrics	Associated esophageal body contractile responses	Clinical impression
Normal No obstructive structural or motor explanation for symptoms	Normal EGJ opening (NEO) EGJ-DI ≥ 2.0 mm ² /mm Hg and Maximum EGJ diameter ≥ 16 mm	Normal contractile response Multiple (>1) distinct antegrade contractions	A major, actionable motor disorder is unlikely
Obstruction Findings support structural or motor obstruction in the presence of compatible symptoms and endoscopy findings	Reduced EGJ opening (REO) EGJ-DI < 2.0 mm ² /mm Hg and Maximum EGJ diameter < 12 mm	Absent contractile response No esophageal body contractions	Nonspastic achalasia is likely
		Diminished contractile response Criteria not fulfilled for spastic, normal, or absent contractile response < 40 mm Hg pressure increase	
		Spastic contractile response Sustained LES contraction (sLESc) Sustained occluding contraction (SOC)	Spastic achalasia is likely
		Disordered contractile response Criteria not fulfilled for spastic, normal, or absent contractile response > 40 mm Hg pressure increase	
Inconclusive Findings are clinically inconclusive, and additional evaluation is necessary to define mechanism for symptoms	Inconclusive EGJ opening (IEO) Isolated EGJ-DI < 2.0 mm ² /mm Hg or Isolated maximum EGJ diameter < 16 mm, but not REO	Normal contractile response	Mechanical obstruction is possible
		Any contractile response pattern	Motor and mechanical obstruction remain possible

NOTE. EGJ metrics drive the final diagnosis of obstruction or no obstruction, and esophageal body contractile responses can provide additional characterization of secondary peristalsis.

DI, distensibility index; EGJ, esophagogastric junction; FLIP, functional lumen imaging probe; IEO, inconclusive EGJ opening; LES, lower esophageal sphincter; NEO, normal EGJ opening; REO, reduced EGJ opening; sLESc, >5 s LES diameter reduction with increase in pressure; SOC, >10 s nonpropagating lumen occluding contraction with increase in pressure.

In patients with obstructive esophageal symptoms when index endoscopy does not demonstrate a mechanism for symptoms, FLIP may be considered if equipment and interpretation expertise are readily available. FLIP can be cost-effective during index endoscopy when it is completely normal or when it facilitates early achalasia diagnosis, while adding an average of only 5 additional minutes to the procedure time.^{25–27} When the endoscopic examination is not suggestive of achalasia, finding reduced or inconclusive EGJ opening may prompt re-inspection for a subtle stricture or ring that could respond to dilation, however, FLIP is typically performed prior to dilation.¹ Spastic contractile response patterns can represent a primary process or be secondary to anatomic factors, such as hiatus hernia or mucosal inflammation (eg, EoE). FLIP remains an option to assess esophageal motility in patients when the HRM catheter fails to traverse the LES, when discomfort or gagging precludes an adequate HRM study, or when alternate testing demonstrates a dilated or tortuous esophagus, predicting HRM placement could fail.²⁸

Best Practice Advice 3: Clinicians may use common sedation options for endoscopy (ie, propofol, fentanyl, and/or midazolam), as these medications have not been demonstrated to have a clinically significant impact on FLIP findings.

Because chronic opiate use has been associated with opiate-induced esophageal dysmotility on HRM, there has been concern regarding opiate administration as part of sedation during endoscopy, although HRM evidence for opiate-induced esophageal dysmotility with acute opiate use is lacking.^{29,30} FLIP and HRM diagnoses were not significantly different in a study that compared FLIP performed with conscious sedation (midazolam and fentanyl) in 174 patients and with monitored anesthesia care (ie, propofol, used with or without fentanyl) in 280 patients.³¹ However, sevoflurane, an inhalation anesthetic used for general anesthesia, can result in higher EGJ diameter and EGJ-DI readings compared with propofol, and can impair secondary peristalsis, likely from more potent neuromuscular blockade.³² It is unknown whether other inhalation anesthetics, such as desflurane and isoflurane, impact FLIP measurements. Clinicians should avoid

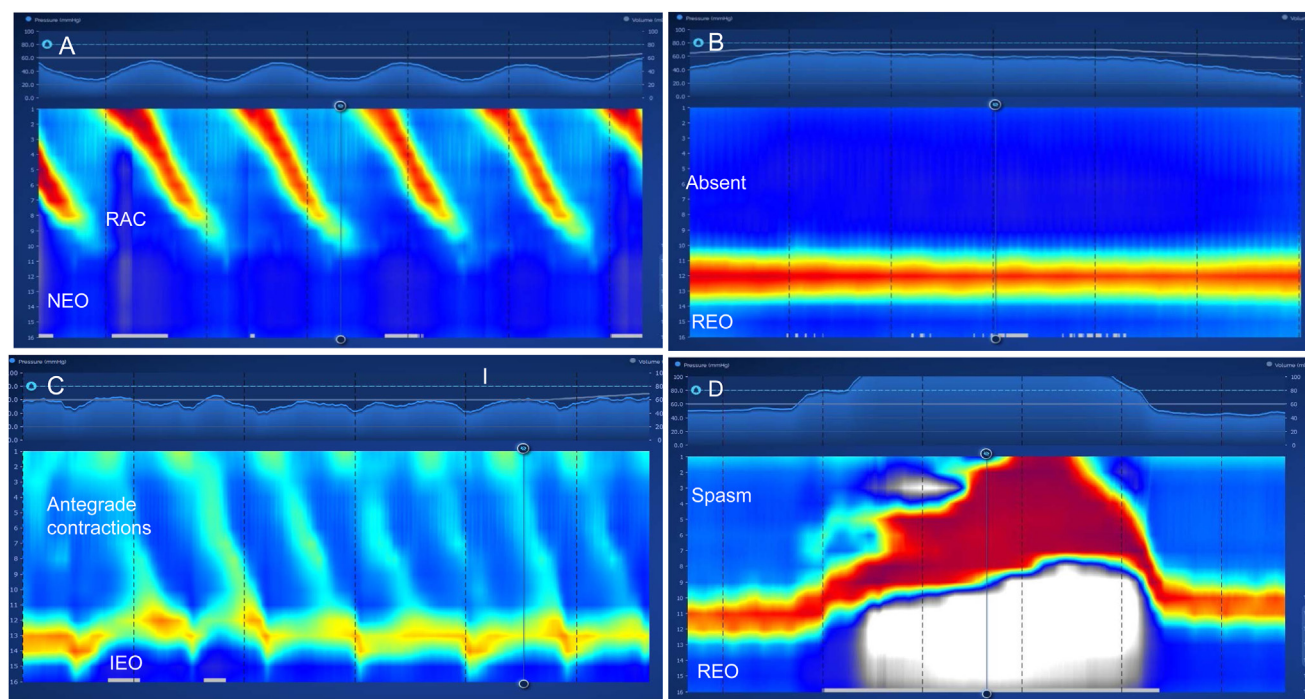


Figure 1. EGJ and esophageal body contractile patterns on FLIP panometry. The EGJ is categorized using DI and maximal EGJ diameter into NEO (A) ($DI \geq 2 \text{ mm}^2/\text{mm Hg}$ and diameter $\geq 16 \text{ mm}$), REO (B, D) ($DI < 2 \text{ mm}^2/\text{mm Hg}$ and diameter $< 12 \text{ mm}$), and IEO (C) ($DI < 2 \text{ mm}^2/\text{mm Hg}$ or diameter $< 16 \text{ mm}$ but not REO). Esophageal body contractile response can be normal when repetitive antegrade contractions (RACs) (A) or multiple antegrade contractions (C) are seen. Other contractile patterns include absent contractile response (B) or spasm (D). Esophageal body contractile patterns complement EGJ categories in determining FLIP panometry diagnosis and planning intervention (see Table 1 and Figure 2).

inhalation anesthetics when assessing esophageal motility with FLIP panometry until more data are available.

FLIP in the Evaluation of Esophageal Motility Disorders

Best Practice Advice 4: NEO on FLIP has a high negative predictive value for disorders of EGJOO on HRM.

A study of 687 symptomatic patients without mechanical obstruction on endoscopy found NEO (Table 1) in $<1\%$ of 243 patients with HRM diagnoses of achalasia or conclusive EGJOO confirmed with TBE.¹⁸ Another study of 184 patients showed none of the 23 patients with HRM-defined achalasia had NEO on FLIP.³³ Furthermore, among 17 patients with EGJOO based only on HRM and NEO on FLIP, none had a $\geq 5\text{-cm}$ barium column height after 5 minutes on TBE, and 9 of 10 patients with longitudinal symptomatic evaluation had good clinical outcomes with conservative treatment.^{24,34} Thus, NEO on FLIP offers high negative predictive value (NPV) for actionable EGJ obstruction as a mechanism for symptoms, and further tests will have limited value.

In contrast, using EGJ-DI without maximum EGJ diameter carries a lower NPV to rule out REO. A study using the Standards for Reporting Diagnostic Accuracy Studies approach showed that FLIP diagnostic performance to rule out achalasia or conclusive EGJOO varied according to the EGJ-DI threshold applied, with NPVs of 90% for EGJ-DI $>2.0 \text{ mm}^2/\text{mm Hg}$, 96% for EGJ-DI $>2.8 \text{ mm}^2/\text{mm Hg}$, and 97% for EGJ-DI $>3.0 \text{ mm}^2/\text{mm Hg}$.³⁴

Combining NEO with a normal esophageal body contractile response pattern provides additional evidence against achalasia. In a study of 706 symptomatic patients, none of 108 patients with a normal contractile response—defined as repetitive antegrade contractions—had achalasia on HRM.¹⁰ However, because absent esophageal body contractile response is reported in as many as 23%–43% of patients with normal primary peristalsis on HRM, absence of secondary peristalsis does not necessarily indicate a distinct motor disorder when NEO is identified.^{22,35}

Best Practice Advice 5: If upper endoscopy demonstrates findings strongly supportive of achalasia, clinicians may use FLIP findings to diagnose achalasia.

FLIP findings of REO and abnormal contractile response patterns are strongly associated with achalasia. Prior work demonstrated that 92% of patients ($n = 185$ of 202) with REO and absent or impaired/disordered esophageal body contractile response had achalasia or conclusive (TBE-confirmed) EGJOO on HRM.^{9,18}

A key advantage of FLIP over HRM is the accompanying high-quality endoscopy, which provides valuable pretest probability to complement FLIP findings, including distinction of idiopathic achalasia from pseudoachalasia. Endoscopic features of achalasia include retained food/fluid (esophageal contents), dilated lumen (anatomy), resistance traversing the EGJ, and stasis changes of esophageal mucosa, conveniently quantified as the CARS (content, anatomy, resistance, and stasis) score (Supplementary Table 1).³⁶ A CARS score ≥ 4 by itself had a specificity of 99% and

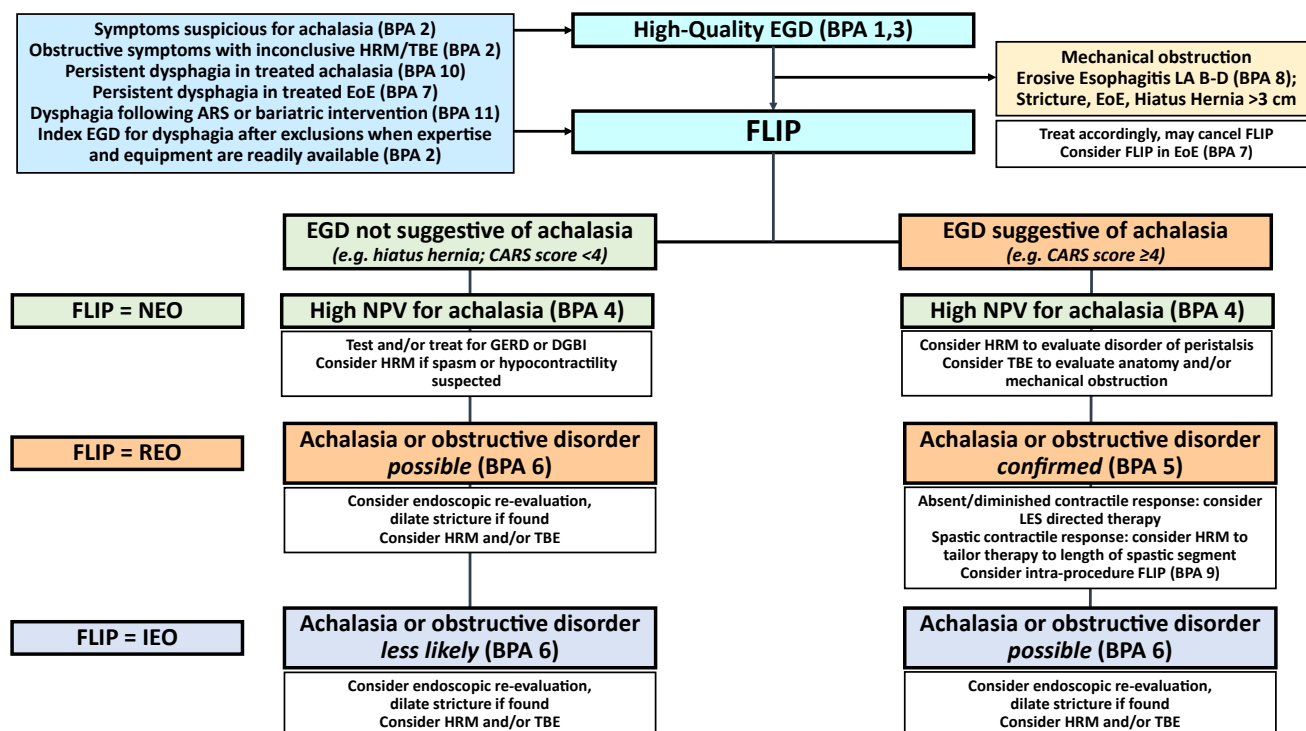


Figure 2. Practical clinical algorithm for utilization and interpretation of FLIP findings. A high-quality esophagogastroduodenoscopy (EGD) evaluates for alternate explanations for obstructive symptoms. The FLIP catheter is introduced transorally in the sedated patient in the left lateral position. FLIP is interpreted in the context of clinical presentation and EGD findings. When EGD is not suggestive of achalasia or outflow obstruction, NEO has high NPV for achalasia. In contrast, when EGD is suggestive of achalasia, REO with absent or reduced contractile function is conclusive for achalasia, and management can proceed without need for additional testing. IEO, spastic contractile response, and EGD-FLIP discordance require alternate diagnostic testing, which can include HRM and/or TBE. The Clinical Practice Update Best Practice Advice statements are represented in this clinical algorithm. CARS score, retained food/fluid (esophageal contents), dilated lumen (anatomy), resistance traversing the EGJ, and stasis changes of esophageal mucosa; DGBI, disorder of gut-brain interaction; LA, Los Angeles esophagitis grade.

sensitivity of 68%–72% to identify achalasia.^{36,37} A CARS score of 0–1 combined with NEO on FLIP had an NPV of 100% for achalasia spectrum disorders and a CARS score ≥ 4 with REO on FLIP had a positive predictive value of 97% for nonspastic achalasia.³⁸ Therefore, achalasia with absent or diminished contractile response can be diagnosed when clinical presentation is complemented with an abnormal CARS score and REO. However, if there is diagnostic uncertainty or if FLIP findings suggest spasm, additional testing using HRM and/or TBE can help clarify the diagnosis prior to pursuing irreversible LES disruption.^{8,9,39}

Best Practice Advice 6: If upper endoscopy does not demonstrate findings supportive of achalasia, abnormal EGJ opening on FLIP should prompt further diagnostic testing.

Although REO is observed in the vast majority of patients with achalasia, it can be seen in 5%–11% of symptomatic patients without HRM disorders of EGJOO.^{9,22,40} Even though the clinical significance of REO on FLIP when HRM EGJ outflow pressures are normal is incompletely understood, esophageal retention on TBE and symptom improvement with EGJ-directed therapies have been reported in carefully selected patients.^{21,33,41} Borderline or

inconclusive EGJ opening was reported in 25%–28% of patients with normal EGJ outflow pressure on HRM and could indicate obstruction.^{1,18,40}

Given the potential for FLIP-HRM discordance, if abnormal EGJ opening on FLIP is not supported by endoscopic evidence of achalasia, additional testing should be pursued before achalasia is diagnosed.^{9,21,22} Careful re-inspection during the same endoscopic encounter can identify strictures or rings contributing to FLIP findings, which can be addressed with endoscopic dilation.¹ If mechanical obstruction is excluded, further testing with HRM and/or TBE should be performed to increase confidence in achalasia diagnosis.

FLIP in the Evaluation of Eosinophilic Esophagitis

Best Practice Advice 7: Clinicians may consider performing FLIP if the mechanism of persisting dysphagia in treated EoE is not apparent on high-quality upper endoscopy or histopathology.

FLIP can assess biomechanical properties of the esophageal wall in EoE. Static esophageal diameter despite

increasing distention volume and pressure, termed *distensibility plateau*, was lower in patients with prior food impaction compared with those with dysphagia alone.⁴² Low DI, and particularly low distensibility plateau, correlates with greater symptom duration and diagnostic delay in EoE.⁴³ Abnormal esophageal body contractile response is associated with more severe fibrostenosis, indicating that EoE disease severity can impact esophageal motor function.⁴⁴ Composite assessment of esophageal body compliance, contractility, EGJ-DI, and maximal diameter has been shown to correlate with higher esophageal eosinophil counts and endoscopic reference scores when abnormal in a cohort of 215 patients with EoE.^{45,46} Thus, FLIP represents a feasible and useful marker for esophageal remodeling, particularly in symptomatic patients.⁴⁶⁻⁴⁸ Rarely, an achalasia-like pattern can be seen in EoE that may be related to fibrostenotic change manifesting as pseudoachalasia,⁴⁹ and irreversible LES-disruptive treatments should be applied cautiously and only if supported by additional complementary tests. FLIP may therefore help explain mechanisms of persistent dysphagia in EoE by identifying luminal restriction, likelihood of food impaction, and superimposed obstructive motility disorders using distensibility plateau, esophageal diameter, and contractile response pattern.

Finally, in a cohort of 46 patients studied before treatment, abnormal composite FLIP assessment scores were associated with histologic PPI nonresponse.^{45,46} Thus, while FLIP should not be used to diagnose EoE, FLIP assessment could potentially help guide optimal therapeutic options and monitor treatment response.

FLIP in the Evaluation of Gastroesophageal Reflux Disease

Best Practice Advice 8: Clinicians should not use FLIP findings to diagnose gastroesophageal reflux disease or to determine the necessity for antireflux intervention.

FLIP studies evaluating EGJ barrier function in gastroesophageal reflux disease (GERD) have not demonstrated a discriminative threshold for EGJ-DI in segregating symptomatic GERD from controls.^{50,51} Whether GERD was defined based on objective reflux metrics on ambulatory reflux monitoring or symptoms alone, DI values varied widely, including within the same ranges as normative controls.⁵⁰⁻⁵⁴ There was no correlation between EGJ-DI values and acid exposure times (AETs) in 2 retrospective evaluations of 164 patients and 69 patients who underwent both FLIP and ambulatory reflux monitoring.^{55,56} Among 18 PPI nonresponders with normal AET on all 4 days of a 96-hour wireless pH monitoring study, the likelihood of DI <2.8 mm²/mm Hg was higher (83.3%) compared with patients with abnormal reflux burden (23.5%),⁵⁶ which is likely a function of FLIP identifying abnormal EGJ opening, rather than a role for FLIP in GERD. Impaired esophageal body contractile response has been associated with abnormal AET in a study with 25 patients,⁵⁷ supporting abnormal esophageal clearance contributing to abnormal AET. Furthermore, FLIP can identify a hiatus hernia, with

84% correlation with HRM-identified hiatus hernia in a study of 100 consecutive patients.⁵⁸ However, FLIP can elicit or enlarge a hiatus hernia in up to 25%; patients with these features trended toward having barium retention on TBE, suggesting a potential obstructive component rather than increased compliance.⁵⁸ Because symptoms and reflux burden primarily determine need for antireflux intervention rather than EGJ compliance or presence of hiatus hernia, FLIP findings alone cannot determine need for antireflux intervention, although FLIP can identify achalasia and obstructive syndromes that refute reflux disease in PPI nonresponders.³⁵

Utility of FLIP in Foregut Interventions

Best Practice Advice 9: When available, clinicians should consider performing FLIP intraoperatively during myotomy (per-oral endoscopic myotomy or laparoscopic Heller myotomy) to guide adequacy of LES disruption.

FLIP has demonstrable value when performed during intervention for achalasia spectrum disorders. By evaluating the LES while per-oral endoscopic myotomy (POEM) or laparoscopic Heller myotomy (LHM) are in progress, intraoperative FLIP can direct myotomy length⁵⁹ and confirm the adequacy of LES disruption in real time,^{59,60} because EGJ-DI reliably improves after myotomy.⁶¹⁻⁶⁵ Data for intraoperative FLIP during pneumatic dilation are less robust, although FLIP has been found to assess adequacy of dilation even in this setting.⁶⁶

Intraoperative FLIP measurements, particularly EGJ-DI, associate with postoperative symptomatic outcomes quantified using Eckardt scores, despite some inconsistent data.⁶⁷ In essence, intraoperative EGJ-DI $<2-3$ mm²/mm Hg may predict persisting symptoms,^{63,68-70} and may prompt additional myotomy.⁷¹⁻⁷⁴ In contrast, high intraoperative EGJ-DI may associate with postoperative reflux and/or erosive esophagitis after LHM⁶² or POEM.^{74,75} Additional data are needed to define the ideal target thresholds for EGJ-DI after LES myotomy, as well as the role of FLIP in guiding myotomy length, such as for type III achalasia.^{71,76}

FLIP in Symptom Evaluation After Foregut Interventions

Best Practice Advice 10: Clinicians may consider performing FLIP in the evaluation of persisting symptoms of esophageal obstruction (ie, dysphagia, esophageal-type regurgitation, and/or meal-related chest pain) after treatment of achalasia spectrum disorders.

FLIP can be used to evaluate patients with persisting esophageal obstructive symptoms after achalasia management.^{2,3} After LES disruption, an EGJ-DI $<2-3$ mm²/mm Hg in symptomatic patients is associated with higher Eckardt scores and suboptimal esophageal emptying, suggesting ineffective or incomplete LES disruption.^{77,78} This may support additional LES-directed therapy.⁷⁹ In contrast, NEO on FLIP should prompt consideration of alternate explanations for symptoms, including abnormal esophageal body anatomy and GERD.^{66,79} Although EGJ-DI values increase

dramatically during or immediately after POEM or LHM, mean EGJ-DI values are lower at 12-month follow-up, but remain higher than preoperative values.⁶²

Best Practice Advice 11: Clinicians may consider performing FLIP in the evaluation of symptoms of esophageal obstruction after invasive foregut intervention (ie, antireflux or bariatric intervention).

Intraoperative FLIP during ARS and endoscopic reflux procedures is feasible, and EGJ-DI has been used to tailor fundoplication in attempts to prevent postoperative dysphagia.^{52,53,80–82} In a study of 33 patients, DI values from intraoperative FLIP affected further surgical methodology, but symptomatic outcomes were not reported.⁸³ In a cohort of 175 patients, EGJ-DI <2 mm²/mm Hg after ARS was associated with dysphagia symptoms.⁸⁴ Patients who developed dysphagia after robotic ARS had greater changes in diameter and DI compared with those without dysphagia.⁸⁵ Tightness of hiatal closure appears to impact DI more than the actual fundoplication.⁸⁶ Ideal EGJ-DI ranges have been proposed for Toupet (EGJ-DI 2.6–3.7 mm²/mm Hg) and Nissen (EGJ-DI >2.2 mm²/mm Hg) funduplications,⁸⁷ with the potential for guiding intraoperative decision making, such as deferring fundoplication, altering the type of ARS, or loosening or tightening the wrap.⁸³ However, in another study using intraoperative FLIP, there were no significant differences in metrics between patients with and without suboptimal postoperative outcomes on follow-up of up to 2 years, even when controlling for type of fundoplication and intraoperative use of bougies.⁸⁸

Data are limited on FLIP performed after other foregut interventions, such as magnetic sphincter augmentation and bariatric surgery.⁸⁹ In 1 study, postoperative diameter and EGJ-DI were lower after magnetic sphincter augmentation compared with ARS, but symptomatic outcomes were similar.⁹⁰ In another study, magnetic sphincter augmentation was associated with higher rates of postoperative dysphagia compared with partial or complete fundoplication, despite similar postoperative EGJ-DI measurements.⁹¹ FLIP performed after Roux-en-Y gastric bypass or sleeve gastrectomy has demonstrated higher secondary dysmotility (most commonly absent contractility) compared with obese patients with native anatomy.^{92,93}

Among a cohort of patients undergoing FLIP for symptomatic esophageal evaluation after ARS, abnormal secondary peristalsis was encountered in 84%, potentially worsening with longer duration post-ARS and/or EGJ outflow resistance.⁹⁴ One study reported the utility of FLIP in evaluating for EGJ obstruction and directing treatment for post-fundoplication dysphagia through dilation, with good symptomatic outcomes.⁹⁵ Therefore, presuming disorders of EGJOO were appropriately ruled out prior to intervention, FLIP can guide management, including consideration of EGJ dilation when symptoms of postoperative esophageal obstruction are confirmed by REO.

Conclusions

Over the past decade and a half, accumulating evidence has demonstrated the diagnostic efficacy of FLIP in

characterizing esophageal obstructive processes, whether performed electively after high-quality endoscopy, intraoperatively during EGJ disruption and ARS, or following suboptimal symptom response after invasive esophageal intervention. FLIP is safe,^{1,8,96} adds minimal time to the standard endoscopic procedure, does not appear to be impacted by standard sedation used during endoscopy, and may provide immediate management options after real-time interpretation. Operator education is paramount, as accurate distinction of reliably obstructive metrics from reactive spastic contractions or normal lumen occlusion helps ensure appropriate intervention. The recently published FLIP Panometry classification scheme standardizes protocols, terminology, and utilization of metrics in the diagnosis of motor and structural esophageal disorders.¹

Future Directions

Similar to other esophageal diagnostic modalities, inconclusive and discordant findings are also encountered with FLIP, and future research is anticipated to further refine next steps in such scenarios. The esophageal smooth muscle response to distention, including the impact of esophageal reactivity and hypersensitivity, warrants further clarification in interpretation of paroxysmal and sustained LES and distal esophageal contractions. As it has with other esophageal diagnostic modalities,⁹⁷ artificial intelligence is expected to further enhance FLIP interpretation, including when FLIP findings are inconclusive.^{39,98} Research is already underway in developing response prediction paradigms, such as in EoE when baseline FLIP characteristics might affect choice of therapy.

Finally, the role of FLIP in adjudicating the diagnosis of functional dysphagia warrants additional research. Future iterations of the FLIP Panometry classification are anticipated as new research uncovers novel diagnostic paradigms.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at [10.1053/j.gastro.2025.05.011](https://doi.org/10.1053/j.gastro.2025.05.011).

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Conflicts of interest

These authors disclose the following: Dustin A. Carlson consults for Medtronic, Diversatek, Braintree, Medpace, Phathom Pharmaceuticals, and Regeneron/Sanofi; and holds a license from Medtronic on FLIP Classifier Model. Amit Patel consults for Medpace, Renexxion, Elsevier, and Sanofi. C. Prakash Gyawali consults for Medtronic, Phathom, Braintree, and Carnot. The remaining author discloses no conflicts.

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Supplementary Table 1. Calculation of the CARS Score^a From High-Quality Endoscopy

Contents	Anatomy	Resistance ^b	Stasis
0 = none	0 = nondilated lumen	0 = no resistance	0 = no stasis changes or <i>Candida</i> esophagitis
1 = liquid content	1 = dilated lumen	1 = mild resistance	1 = chronic mucosal stasis changes
2 = solid content	2 = severely dilated lumen	2 = severe resistance	1 = <i>Candida</i> esophagitis

From Ellison et al,³⁶ adapted with permission.

^aThe CARS score is calculated during a high-quality endoscopy after excluding strictures and other mechanical causes of dysphagia. The sum of the scores for the 4 components (content, anatomy, resistance, and stasis) forms the CARS score (total score: 0–8). The optimal threshold for achalasia is ≥ 4.0 based on receiver operating characteristic analysis, with sensitivity of 68%, specificity of 99%, positive predictive value of 93%, and NPV of 94%. The CARS score serves to inform a pretest probability of achalasia and, when combined with FLIP findings during the same high-quality endoscopy, NPV for achalasia can be as high as 100%, and positive predictive value for nonspastic achalasia as high as 97%.

^bResistance to endoscope passage across the EGJ.